

LATIHAN PRAKTIKUM JOBSHEET 5

Kode main:

```
public class NilaiMahasiswaMain23 {  
    static int maxUTSIndexDC(NilaiMahasiswa23[] arr, int l, int r) {  
        if (l == r) {  
            return l;  
        }  
  
        int mid = (l + r) / 2;  
        int leftIndex = maxUTSIndexDC(arr, l, mid);  
        int rightIndex = maxUTSIndexDC(arr, mid + 1, r);  
  
        return arr[leftIndex].nilaiUTS >= arr[rightIndex].nilaiUTS ?  
leftIndex : rightIndex;  
    }  
  
    static int minUTSIndexDC(NilaiMahasiswa23[] arr, int l, int r) {  
        if (l == r) {  
            return l;  
        }  
  
        int mid = (l + r) / 2;  
        int leftIndex = minUTSIndexDC(arr, l, mid);  
        int rightIndex = minUTSIndexDC(arr, mid + 1, r);  
  
        return arr[leftIndex].nilaiUTS <= arr[rightIndex].nilaiUTS ?  
leftIndex : rightIndex;  
    }  
  
    static double rataRataUASBF(NilaiMahasiswa23[] arr) {  
        int total = 0;  
        for (NilaiMahasiswa23 mhs : arr) {  
            total += mhs.nilaiUAS;  
        }  
        return (double) total / arr.length;  
    }  
  
    public static void main(String[] args) {  
        NilaiMahasiswa23[] mhs = new NilaiMahasiswa23[] {  
            new NilaiMahasiswa23("Ahmad", "220101001", 2022, 78,  
82),
```

```

        new NilaiMahasiswa23("Budi", "220101002", 2022, 85,
88),
        new NilaiMahasiswa23("Cindy", "220101003", 2021, 90,
87),
        new NilaiMahasiswa23("Dian", "220101004", 2021, 76,
79),
        new NilaiMahasiswa23("Eko", "220101005", 2023, 92, 95),
        new NilaiMahasiswa23("Fajar", "220101006", 2020, 88,
85),
        new NilaiMahasiswa23("Gina", "220101007", 2023, 80,
83),
        new NilaiMahasiswa23("Hadi", "220101008", 2020, 82, 84)
    };

    int idxMax = maxUTSIndexDC(mhs, 0, mhs.length - 1);
    int idxMin = minUTSIndexDC(mhs, 0, mhs.length - 1);
    double rataUAS = rataRataUASBF(mhs);

    System.out.println("Nilai UTS tertinggi (DC): " +
mhs[idxMax].nilaiUTS + " - " + mhs[idxMax].nama + " (" +
mhs[idxMax].nim + ")");

    System.out.println("Nilai UTS terendah (DC): " +
mhs[idxMin].nilaiUTS + " - " + mhs[idxMin].nama + " (" +
mhs[idxMin].nim + ")");

    System.out.println("Rata-rata nilai UAS (BF): " + rataUAS);
}
}

```

kode object:

```

public class NilaiMahasiswa23 {
    public String nama;
    public String nim;
    public int tahunMasuk;
    public int nilaiUTS;
    public int nilaiUAS;

    public NilaiMahasiswa23(String nama, String nim, int tahunMasuk,
int nilaiUTS, int nilaiUAS) {
        this.nama = nama;
        this.nim = nim;
        this.tahunMasuk = tahunMasuk;
    }
}

```

```
        this.nilaiUTS = nilaiUTS;  
        this.nilaiUAS = nilaiUAS;  
    }  
}
```

hasil:

Nilai UTS tertinggi (DC): 92 - Eko (220101005)

Nilai UTS terendah (DC): 76 - Dian (220101004)

Rata-rata nilai UAS (BF): 85.375